

PROJECT SUBMISSION: GhostLogin

PROGRAM CODE: NX201

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GhostLogin (NX201) - Project Summary & Execution Flow

This document outlines the technical implementation of the GhostLogin automated SSH assessment tool, designed to validate network exposure and test authorized credential sets in a controlled environment.

1. User Input and Validation The script initializes by prompting the user to define the scope of the assessment by inputting an IP range or subnet (e.g., in CIDR notation). To prevent execution errors, the input is strictly validated using a Regular Expression (Regex) to ensure it is correctly formatted before proceeding.

2. Scanning for SSH Services Once the IP range is validated, the script employs nmap to perform a targeted scan specifically for active SSH services (Port 22). The output is parsed using awk to extract, collect, and list only the IP addresses of the hosts that have an active SSH service running, filtering out closed or unresponsive targets.

3. Credential Brute Forcing For the authentication testing phase, the script is configured to be flexible. It allows the use of either a user-provided credentials file (such as pass.txt) or defaults to a dynamically generated, built-in list of common SSH credentials. Using nested loops, the tool systematically attempts to brute-force SSH logins on the discovered hosts using these parsed credentials.

4. Post-Exploitation Automation Upon a successful login sequence, the script demonstrates post-access verification by running a predefined command on the remote machine non-interactively. To maintain full automation, the script utilizes sshpass to handle SSH password passing automatically without interactive prompts.

, and includes the `-o StrictHostKeyChecking=no` flag to bypass host key verification prompts. The tool securely creates a hidden file (`.ghostlogin_proof`) as a proof of concept without opening an interactive shell.

5. Output and Reporting Throughout the execution, the script provides real-time terminal output, displaying exactly which IPs were successfully accessed, the valid credentials discovered, and confirming where the post-exploitation command was successfully executed.

6. Creativity & Intentional Upgrade To fulfill the project's creativity requirement, I implemented an **Automated HTML Report Generation** feature. Instead of solely relying on terminal output, the script dynamically writes the assessment results into a structured, styled HTML file (`GhostLogin_Report.html`). This intentional improvement enhances the tool's reporting capabilities and usability by providing a clean, exportable format (easily printable to PDF) that visually distinguishes between successful and failed access attempts.



